

* We can think of the superconf. construction as the "scaffolding" needed to construct a physical theory. In particular, we can remove its "origin" and work with the physical theory we obtain from it.

Elements of Gauged Sugra:

(i) The choice of a set of chiral mults, $z^a, \bar{z}^{\bar{a}}$, whose kinetic terms are determined by the Kähler potential $K(z, \bar{z})$. The metric of the scalar target space is invariant under Kähler trans.

$$g_{a\bar{b}} = \partial_a \partial_{\bar{b}} K(z, \bar{z}) \quad | \quad K \rightarrow K + f + \bar{f}$$

(ii) Choice of gauge multiplets (A_m^A, λ^A, D^A) . The Gauge mult. kinetic terms are determined by holo $\int d^4x \quad f_{AB}(z) = f_{BA}(z)$

(iii) Choose a holo superpotential $\bar{W}(z)$. The scalar potential $V(z, \bar{z})$, fermion mass terms and couplings depend on $W(z)$ and $\bar{W}(\bar{z})$

$$\begin{aligned} W(z) &\mapsto W(z) \quad (\text{coordinate}) \\ W(z) &\mapsto e^{-\kappa^2 f(z)} W(z) \quad (\text{Kähler}) \end{aligned}$$

(iv) Lie Group G w/ structure constants f_{AB}^C provided that K, f_{AB} and W satisfy sufficient conditions.

* Generators are a subset associated w/ (A_m^A, λ^A, D^A) w/

gauge parameter θ^A . $\delta z^A = \theta^A K_A^A(z)$

determined by the moment maps $P_A(z, \bar{z})$
 $K_A^A(z) = -i g^{\alpha\bar{\beta}} \partial_{\bar{\beta}} P_A \quad | \quad \nabla_\alpha \partial_P P_A = 0$

$$K_A^A g_{\alpha\bar{\beta}} K_B^{\bar{\beta}} - K_B^A g_{\alpha\bar{\beta}} K_A^{\bar{\beta}} = i f_{AB}^C P_C$$

Kähler potential run charge via:

$$\delta K = \theta^A (r_A - \bar{r}_A) \quad , \quad r_A = K_A^a \partial_a K + i P_A$$

$$\Rightarrow \underline{W_a K_A^a + K^2 r_A W = 0}$$

The physical Action:

* Putting together all the ingredients:

$$\begin{aligned} e^{-1} \mathcal{L} = & \frac{1}{2\kappa^2} [R(e) - \bar{\Psi}_m R^m] \cdot g_{\alpha\bar{\beta}} [\hat{\partial}_m z^\alpha \hat{\partial}_m \bar{z}^\beta + \frac{1}{2} \bar{\chi}^\alpha \not{D}^{(0)} \chi^\beta + \frac{1}{2} \bar{\chi}^\beta \not{D}^{(0)} \chi^\alpha] \\ & - V + \text{Re}(f)_{AB} \left[-\frac{1}{4} F_{\mu\nu}^A F^{\mu\nu B} - \frac{1}{2} \lambda^A \not{D}^{(0)} \lambda^B \right] \\ & + \frac{1}{4} i \left[\text{Im}(f)_{AB} F_{\mu\nu}^A F^{\mu\nu B} + (\hat{\partial}_m \text{Im}(f)_{AB}) \bar{\lambda}^A \gamma_\mu \lambda^B \right] \\ & + \frac{1}{8} \text{Re}(f)_{AB} \bar{\Psi}_m \gamma^{\alpha\beta} (F_{\alpha\beta}^A + \hat{F}_{\alpha\beta}^A) \gamma^\mu \lambda^B + \left[\frac{1}{16} g_{\alpha\bar{\beta}} \bar{\Psi}_m \hat{\partial}_m \bar{z}^\beta \gamma^\mu \chi^\alpha + \text{h.c.} \right] \\ & + \left[\frac{1}{4\sqrt{2}} f_{AB\alpha} \bar{\lambda}^A \gamma^{\alpha\beta} \hat{F}_{\alpha\beta}^B \chi^\alpha + \text{h.c.} \right] + \mathcal{L}_m + \mathcal{L}_{\text{mix}} + \mathcal{L}_{\text{up}} \end{aligned}$$

where

$$R^m := \gamma^{\mu\rho\sigma} \left(\partial_\rho + \frac{1}{4} \omega_\rho^{ab} \gamma_{ab} - \frac{3}{2} i \not{A}_\rho \gamma_5 \right) \Psi_\sigma$$

$$\hat{\partial}_m z^\alpha := \partial_m z^\alpha - A_m^A K_A^\alpha$$

$$\not{D}_m^{(0)} \chi^\alpha = \left(\partial_m + \frac{1}{4} \omega_m^{ab} \gamma_{ab} + \frac{3}{2} i \not{A}_m \right) \chi^\alpha - A_m^A \frac{\partial K_A^\alpha}{\partial z^\beta} \chi^\beta + \Gamma_{\beta\gamma}^\alpha \chi^\gamma \hat{\partial}_m \bar{z}^\beta$$

$$\Gamma_{\beta\gamma}^\alpha = g^{\alpha\bar{\delta}} \partial_\beta g_{\delta\gamma}$$

$$\begin{aligned} \mathcal{L}_m &= \frac{1}{6} i \kappa^2 (\partial_m z^A \partial_m K - \partial_m \bar{z}^{\bar{A}} \partial_m \bar{K}) - \frac{1}{3} \kappa^2 A_m^A P_A \\ &= \frac{1}{6} i \kappa^2 [\hat{\partial}_m z^A \partial_m K - \hat{\partial}_m \bar{z}^{\bar{A}} \partial_m \bar{K}] + A_m^A (r_A - \bar{r}_A) \end{aligned}$$

where $\not{A}_m$ transforms as

$$\delta \mathcal{L}_h = -\frac{1}{3} k^2 \partial_\mu (\theta^A \Gamma_A + \text{Im}(f))$$

under gauge + Kähler

- Recall the scalar potential has a pos. and neg. definite elements: $V = V^+ + V^-$

$$V_- := -3k^2 e^{\kappa^2 K} \bar{W} W$$

$$V_+ = e^{\kappa^2 K} \nabla_a W g^{a\bar{b}} \bar{\nabla}_{\bar{b}} \bar{W} + \frac{1}{2} (\text{Re}(f))^{AB} P_A P_B$$

and W appears wr its Kähler cov. derivative

$$\nabla_a W = \partial_a W + \kappa^2 \partial_a K W$$

$$D_\mu^{(0)} \lambda^A = (\partial_\mu + \frac{1}{4} \omega_\mu^{ab} \gamma_{ab} - \frac{3}{2} i \mathcal{A}_\mu \gamma_5) \lambda^A - A_\mu^C \lambda^B f_{BC}^A$$

- We have terms of the form $\bar{\Psi}_\mu \mathcal{J}^\mu$

Here $\mathcal{J}$ is the basic supercurrent of the gauge and chiral multiplets.

*The supercovariant gauge curvature $\hat{F}_{ab}$:

$$\hat{F}_{ab}^A = e_a^\mu e_b^\nu (\gamma \partial_{[\mu} A_{\nu]}^A + f_{BC}^A A_\mu^B A_\nu^C + \bar{\Psi}_{[\mu} \gamma_{\nu]} \lambda^A)$$

- $(\bar{F} + \hat{F})$ is used to ensure that the symm of the theory are realised at the level of the E.O.M.

Fermionic mass Lagrangian:

$$\mathcal{L}_m = \frac{1}{2} m_{3/2} \bar{\Psi}_\mu P_R \gamma^{\mu\nu} \Psi_\nu - \frac{1}{2} M_{AB} \bar{\chi}^A \chi^B - m_{\lambda A} \bar{\chi}^A \lambda^A - \frac{1}{2} M_{AB} \bar{\lambda}^A P_L \lambda^B + \text{h.c.}$$

To obtain these Recall this was originally

$$\mathcal{L}_m = \frac{1}{2} W \bar{\Psi}_\mu P_R \gamma^{\mu\nu} \Psi_\nu - \frac{1}{2} \nabla_\mu W \bar{\Omega}^\mu \Omega^\mu + \frac{1}{4} G^{\mu\nu} \bar{W}_\mu f_{AB\mu} \bar{\lambda}^A P_L \lambda^B$$

$$+ \sqrt{2} i \left[-\partial_\mu P_A + \frac{1}{4} f_{AB\mu} (\text{Re}(f))^{BC} P_C \right] \lambda^A \Omega^\mu + \text{h.c.}$$

$$\rightarrow W = \gamma^3 \bar{W} = e^{\kappa^2 K/2} \bar{W}$$

$$\rightarrow \nabla_\mu W \bar{\gamma}^\mu \bar{\chi}^\alpha \bar{\nabla}_\alpha Z^I \gamma^\mu \chi^\beta \nabla_\beta Z^J = \gamma^3 \nabla_\mu W$$

$$\rightarrow f_{AB\mu} = \nabla_\mu f_{AB} = \gamma f_{AB\mu} \nabla_\mu Z^I$$

$$M_{1/2} = K^2 e^{u^2 K/2} \bar{W}(z)$$

$$m_{\alpha\beta} = e^{u^2 K/2} \nabla_\alpha \nabla_\beta \bar{W} \equiv e^{u^2 K/2} [\partial_\alpha + (K^2 \partial_\alpha K)] \nabla_\beta W - e^{u^2 K/2} \Gamma_{\alpha\beta}^{\gamma} \nabla_\gamma W$$

$$m_{\alpha A} = i\sqrt{2} [\partial_\alpha P_A - \frac{1}{4} f_{AB\alpha} (Re(f)^{-1})^{BC} P_C] = m_{A\alpha}$$

$$M_{AB} = -\frac{1}{2} e^{u^2 K/2} f_{AB\alpha} g^{\alpha\bar{\beta}} \nabla_{\bar{\beta}} W$$

Terms which mix the gravitino & spin 1/2 ferm. are:

$$\mathcal{L}_{mix} = (\bar{\psi} \cdot \gamma) \left[\frac{1}{2} \epsilon_P \lambda^A P_A + \frac{1}{\sqrt{2}} \chi^\alpha e^{u^2 K/2} \nabla_\alpha W \right] + h.c.$$

* For completeness we have the four-fermion terms + $\mathcal{L}_{SO-torsion}$

Claim:

$$\partial_\alpha V = e^{u^2 K/2} (-2\bar{m}_{1/2} \nabla_\alpha W + m_{\alpha\beta} g^{\beta\bar{\gamma}} \bar{\nabla}_{\bar{\gamma}} \bar{W}) - \frac{i}{\sqrt{2}} P_A (Re(f)^{-1})^{AB} m_{\alpha\beta}$$

which is quite important since $\partial_\alpha V = 0$ determines the possible vacuum states of the theory.

Proof: Recall: $[\partial_\alpha, \bar{\nabla}_{\bar{\beta}}] V = e^{-i} (\omega_- - \omega_+) g_{\alpha\bar{\beta}} V$

$$\partial_\alpha V_- = \partial_\alpha (-3K^2 e^{u^2 K} W \bar{W}) = -3K^2 e^{u^2 K} \partial_\alpha K - 3K^2 e^{u^2 K} \partial_\alpha (W \bar{W})$$

$$\partial_\alpha V_- = -3K^2 e^{u^2 K} \partial_\alpha K W \bar{W} - 3K^2 e^{u^2 K} (\partial_\alpha W \bar{W} + W \partial_\alpha \bar{W})$$

$$= -3K^2 e^{u^2 K} \partial_\alpha K W \bar{W} - 3K^2 e^{u^2 K} \partial_\alpha W \bar{W}$$

$$= -3K^2 e^{u^2 K} (\partial_\alpha W + K^2 \partial_\alpha K W) \bar{W} = -3K^2 e^{u^2 K} \nabla_\alpha W \bar{W}$$

$$\partial_\alpha V_+ = \partial_\alpha (e^{u^2 K} \nabla_\gamma W g^{\gamma\bar{\beta}} \bar{\nabla}_{\bar{\beta}} \bar{W} + \frac{1}{2} (Re(f)^{-1})^{AB} P_A P_B)$$

$$\rightarrow K^2 e^{u^2 K} \partial_\alpha K \nabla_\gamma W g^{\gamma\bar{\beta}} \bar{\nabla}_{\bar{\beta}} \bar{W} + e^{u^2 K} \partial_\alpha \nabla_\gamma W g^{\gamma\bar{\beta}} \bar{\nabla}_{\bar{\beta}} \bar{W} + e^{u^2 K} \nabla_\gamma W \partial_\alpha g^{\gamma\bar{\beta}} \bar{\nabla}_{\bar{\beta}} \bar{W} + e^{u^2 K} \nabla_\gamma W g^{\gamma\bar{\beta}} \partial_\alpha \bar{\nabla}_{\bar{\beta}} \bar{W}$$

$$\rightarrow \partial_\alpha \bar{\nabla}_{\bar{\beta}} \bar{W} \equiv e^{-i} (3) g_{\alpha\bar{\beta}} \bar{W}$$

$$= e^{K^2 K} (\nabla_\alpha W) a^{-1}(3) \bar{W} = K^2 e^{K^2 K} \nabla_\alpha W \bar{W}$$

$$\rightarrow g_{\gamma\bar{\kappa}} g^{\bar{\alpha}\beta} = \delta_\gamma^\beta$$

$$\partial_\alpha g_{\delta\bar{\beta}} g^{\bar{\alpha}\gamma} + \partial_\alpha g^{\gamma\bar{\beta}} g_{\bar{\beta}\delta} = 0$$

$$- \partial_\alpha g_{\delta\bar{\beta}} g^{\bar{\beta}\gamma} = g_{\bar{\beta}\delta} \partial_\alpha g^{\gamma\bar{\beta}}$$

$$- g^{\delta\bar{\alpha}} (\partial_\alpha g_{\delta\bar{\beta}}) g^{\bar{\beta}\gamma} = \partial_\alpha g^{\gamma\bar{\alpha}}$$

$$\therefore \partial_\alpha g^{\gamma\bar{\beta}} = - g^{\delta\bar{\beta}} (\partial_\alpha g_{\delta\bar{\alpha}}) g^{\bar{\alpha}\gamma}$$

$$- e^{K^2 K} \nabla_\gamma W g^{\delta\bar{\beta}} \Gamma_{\alpha\delta}^{\gamma\bar{\gamma}} \bar{\nabla}_{\bar{\beta}} \bar{W}$$

$$= K^2 e^{K^2 K} \partial_\alpha K \nabla_\gamma W g^{\delta\bar{\beta}} \bar{\nabla}_{\bar{\beta}} \bar{W} + e^{K^2 K} (\partial_\alpha \nabla_\gamma W) g^{\delta\bar{\beta}} \bar{\nabla}_{\bar{\beta}} \bar{W} \\ - e^{K^2 K} \nabla_\gamma W g^{\delta\bar{\beta}} \Gamma_{\alpha\beta}^{\gamma\bar{\gamma}} \bar{\nabla}_{\bar{\beta}} \bar{W} + K^2 e^{K^2 K} \nabla_\alpha W \bar{W}$$

$$= e^{K^2 K} g^{\beta\bar{\beta}} \bar{\nabla}_{\bar{\beta}} \bar{W} (\partial_\alpha \nabla_\beta W + K^2 \partial_\alpha K \nabla_\beta W - \Gamma_{\alpha\beta}^{\gamma\bar{\gamma}} \nabla_\gamma W)$$

$$= e^{K^2 K/2} g^{\beta\bar{\beta}} \bar{\nabla}_{\bar{\beta}} \bar{W} M_{\alpha\beta} \quad \checkmark$$

$$\rightarrow \partial_\alpha \left(\frac{1}{2} (\text{Re}(f)^{-1})^{AB} P_A P_B \right)$$

$$\frac{1}{2} \partial_\alpha (\text{Re}(f)^{-1})^{AB} P_A P_B + (\text{Re}(f)^{-1})^{AB} P_A \partial_\alpha P_B$$

$$(f^{-1})^{AB} f_{BC} = \delta^A_C$$

$$\partial_\alpha (f^{-1})^{AB} = - f^{AC} \partial_\alpha f_{CD} f^{DB}$$

$$\text{Re}(f)^{AB} = \frac{1}{2} (f^{AB} + \bar{f}^{AB})$$

$$\partial_\alpha (\text{Re}(f)^{AB}) = \frac{1}{2} \partial_\alpha f^{AB} = - \frac{1}{2} f^{AC} \partial_\alpha f_{CD} f^{DB}$$

Not sure about this part: expand

$$f^{AB} = \text{Re}(f^{AB}) + i \text{Im}(f^{AB})$$

Keep Real part:

$$\partial_\alpha (\text{Re}(f)^{-1})^{AB} = - \frac{1}{2} (\text{Re}(f)^{-1})^{AC} \partial_\alpha f_{CD} (\text{Re}(f)^{-1})^{DB}$$

Q: Not sure why $\partial_\alpha f_{CD}$ is not expanded or found

$$\begin{aligned}
&= -\frac{1}{4} \text{Re}(f)^{-1AB} \partial_a f \partial_b (\text{Re}(f)^{-1})^{DC} P_A P_C \\
&\quad + \text{Re}(f)^{-1AB} P_A \partial_a P_B \\
&= (\text{Re}(f)^{-1})^{AD} \left[P_A \partial_a P_B - \frac{1}{4} \partial_a f \partial_b (\text{Re}(f)^{-1})^{DC} P_C \right] \\
&= -\frac{i}{2} \text{Re}(f)^{AB} P_A m_{AB}
\end{aligned}$$

Adding these together we arrive at the above Potential. $\square$

* Recall, we have the Kähler invariant $\mathcal{L}^H$
 $\mathcal{G} = K^2 K + \log(K^6 W \bar{W}) \quad | \quad G_{\alpha\bar{\beta}} = K^2 g_{\alpha\bar{\beta}}$

Claim: $V_F = K^{-4} e^{\mathcal{G}} (G_{\alpha} G^{\alpha\bar{\beta}} G_{\bar{\beta}} - 3)$

Proof:

$$\begin{aligned}
\text{RHS} &= K^{-4} e^{K^2 K} K^6 \bar{W} \bar{W} \left(K^2 \partial_{\alpha} K + \frac{1}{K^6} \frac{\partial_{\alpha} W}{W} \right) \left(\frac{1}{K^2} g^{\alpha\bar{\beta}} \right) \left(K^2 \partial_{\bar{\beta}} K + \frac{1}{K^6} \frac{\partial_{\bar{\beta}} \bar{W}}{\bar{W}} \right) \\
&= e^{K^2 K} \bar{W} \bar{W} \left[\left(K^2 \partial_{\alpha} K + \frac{\partial_{\alpha} W}{W} \right) \left(K^2 \partial_{\bar{\beta}} K + \frac{\partial_{\bar{\beta}} \bar{W}}{\bar{W}} \right) g^{\alpha\bar{\beta}} - 3 \right] \\
&= e^{K^2 K} \left[(K^2 \partial_{\alpha} K W \bar{W} + \partial_{\alpha} W \bar{W}) g^{\alpha\bar{\beta}} W (\partial_{\bar{\beta}} K \bar{W} + \partial_{\bar{\beta}} \bar{W}) - 3 W \bar{W} e^{2K} \right] \\
&= e^{K^2 K} \left[(\nabla_{\alpha} W) g^{\alpha\bar{\beta}} (\bar{\nabla}_{\bar{\beta}} \bar{W}) - 3 \right] \quad \square
\end{aligned}$$

Transformation Rules: (Excluding Lorentz)

$$\delta e_m^a = \frac{1}{2} \bar{\epsilon} \gamma^a \psi_m$$

$$\begin{aligned}
\delta P_L \psi_m &= (\partial_m + \frac{1}{4} \omega_m^{ab} (e) \gamma_{ab} - \frac{3}{2} i \gamma_5) P_L \epsilon + \frac{1}{2} K^2 \bar{\psi}_m e^{K/2} W P_R \epsilon \\
&\quad + \frac{1}{4} K^2 P_L \psi_m \theta^4 (\bar{r}_A - r_A) + (\text{cubic in fermions})
\end{aligned}$$

$$\delta z^a = \frac{1}{2} \bar{\epsilon} \gamma^a \chi + \theta^4 K_A^a$$

$$\begin{aligned}
\delta \chi^a &= \frac{1}{2} P_L \left(\not{\partial} z^a - e^{K^2 K/2} g^{\alpha\bar{\beta}} \bar{\nabla}_{\bar{\beta}} \bar{W} \right) \epsilon \\
&\quad + \theta^4 \left[\frac{\partial K_A}{\partial z^a} \chi^B + \frac{1}{4} K^2 (r_A - \bar{r}_A) \chi^a \right] + (\text{cubic in fermions})
\end{aligned}$$

$$\delta A_m^A = -\frac{1}{2} \bar{\epsilon} \gamma_m \lambda^A + \partial_m \theta^4 + \theta^4 A_m^B f_{BC}^A$$

$$\delta \lambda^A = \left[\frac{1}{4} \gamma^{mn} F_{mn}^A + \frac{1}{2} i \gamma_5 P_L (e) \gamma^A \gamma^B P_R \right] \epsilon$$

+ $\Theta^B [\lambda^C f_{CB}^A + \frac{1}{4} \kappa^2 \chi (\bar{\tau}_D - \tau_D) \lambda^A] + (\text{cubic in ferm})$
 "cubic in fermions": $\sim \chi^a \bar{\chi}^b \bar{\chi}^c \chi^d \in$. These terms
 are omitted because they are not needed.

Kähler:

$$\begin{aligned} P_L \psi_m &\mapsto e^{-i\kappa^2 \text{Im}(f)/2} P_L \psi_m \\ P_L \lambda^A &\mapsto e^{-i\kappa^2 (\text{Im}(f)/2)} P_L \lambda^A \\ \chi^\alpha &\mapsto e^{i\kappa^2 \text{Im}(f)/2} \chi^\alpha \end{aligned}$$

Why? Recall that

$$\delta_T \chi^\alpha = -\frac{3}{2} i \lambda_T \chi^\alpha$$

$$\text{and } \lambda_T = \left(1 - \frac{f}{a} \text{Im}(f) \right)$$

$$\therefore \delta_{\text{Kähler}} \chi^\alpha = \frac{3}{2} \frac{i}{a} \text{Im}(f) \chi^\alpha, \quad a = \frac{3}{\kappa^2}$$

$$\delta_{\tilde{a}} \chi^\alpha = \frac{i\kappa}{2} \text{Im}(f) \chi^\alpha$$

• For $P_L \psi_m$: $\delta \psi_m = \frac{3}{2} i \lambda_T \delta_\chi \psi_m$

$$\therefore \delta_T P_L \psi_m = \frac{3}{2} i \lambda_T P_L \psi_m \Rightarrow$$

$$\delta_{\tilde{a}} P_L \psi_m = -\frac{\kappa^2}{2} i \text{Im}(f) P_L \psi_m.$$

* Recall, we could obtain V , the scalar potential,
 from the sum of squares from the scalar part
 of the transf of physical fermions; so called
 "fermion shift" terms:

$$\delta_S \chi^\alpha = -\frac{1}{\sqrt{2}} e^{\kappa^2 K/2} g^{\alpha\bar{\beta}} \bar{\nabla}_{\bar{\beta}} \bar{W} \quad \parallel \quad \delta_S P_L \lambda^A = \frac{1}{2} i (M_C(M)^{-1})^{AB} P_B$$

$$V = -\frac{1}{2} (D-2)(D-1) \kappa^2 e^{\kappa^2 K} |W|^2$$

$$+ 2 (\delta_S P_L \chi^\alpha) g_{\alpha\bar{\beta}} (\delta_S P_L \bar{\chi}^{\bar{\beta}}) + 2 (\delta_S P_L \lambda^A) \text{Re}(M_{AB}) (\delta_S P_L \lambda^B)$$

Note, the gravitino term gives rise to $V =$

from $\ln(1 - \epsilon) \approx -\epsilon$ $\Rightarrow (1 - \epsilon)(1 - \epsilon) \approx 1 - 2\epsilon$

Further remarks:

z^a : Take to be dimensionless. This makes sense

$$\text{Since } z^a_{(F)} = x^a / x^F$$

$$\therefore g_{\alpha\bar{\beta}} \partial_\alpha z^a \partial_{\bar{\beta}} \bar{z}^{\bar{b}} \Rightarrow g_{\alpha\bar{\beta}} : [M]^2 \Rightarrow K : [M^2]$$

$$K = K^{-2} z^a \delta_{a\bar{b}} \bar{z}^{\bar{b}}$$

$$K : [M]^2, \quad g_{\alpha\bar{\beta}} : [M]^2, \quad W : [M]^3, \quad K : [M]^{-1}$$

$$A_\mu : [M], \quad \gamma : [M]^2, \quad P_A : [M]^2, \quad f_{AB} : [M]^0$$

▷ This is due to the fact that 'metric' terms are often of the form

$$g_{\mu\nu} \partial_\mu \phi^\alpha \partial_\nu \phi^\beta \quad (\text{kinetic term})$$

The kin. term for Ψ :

$$\Psi_\mu \gamma^{\mu\rho} \partial_\rho \Psi$$

$\therefore (\gamma^{\mu\rho} \partial_\rho)$: acts as a metric

Not sure what happens to the remaining $\mathcal{L}$ term.

